AS YOU SIT DOWN

Get your workspace open

FOUR THINGS

- VS Code open on your **ECON209** folder
- Your **README.md** from Lab 1
- Your Problem Set 1 notebook, however far you have got
- Your AI assistant signed in beside it

IF SOMETHING ISN'T WORKING

- Sit next to someone whose setup runs
- Follow along now, fix yours after

APPLIED ECONOMETRICS · RICE UNIVERSITY

Lab Three

Working with AI on code you already have: getting it done, and understanding how it got done.

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Questions, any time

Sewall 207
Mondays

START HERE

Your README.md

```
# ECON 209 — your name
## The course
Applied Econometrics, Fall 2026
Prof. Jeff Dominitz
Using data and assumptions
to make predictions

## What's in this folder
ECON209/
├── README.md       this file
├── decisionlog.md  what and why
├── homework/       private
└── labs/           weekly labs
    (both hold data/ code/ output/)
```

```
## AI use in this class
Yes: writing and debugging code
No:  interpreting results
No:  non-coding questions
No:  anything on an exam

## Our rules for working with AI
1. Work in rounds: read, simplify,
   annotate, question, check
2. Keep the code simple enough
   to explain every line
3. Comment every line I keep
4. Check against something I know
5. I stand by what results mean
```

TONIGHT

Two goals, not equal weight

1 · TASK COMPLETION

- The code runs
- The tables print
- Every question has something under it

2 · DEEP UNDERSTANDING · WEIGHTED MORE

- What each line does
- What it assumes
- Why the number is right
- How you would know if it were wrong

Complete work may not be correct work. Tonight is mostly about the second goal.

HOW WE WORK WITH AN AGENT

Work in rounds

1. **Read.** Have it walk you through the code
2. **Simplify.** The shortest version that does the same thing
3. **Annotate.** A comment on every line
4. **Question.** What does it assume?
5. **Check.** Against something you already know

An agent can run a cell, change one thing, and run it again. A round is done when you can explain the cell, not when it runs.

DEMO · LAB3-PRICE-ANALYSIS.IPYNB

A notebook that is complete

- The same kinds of tables and simulations as Problem Set 1, on the lab's variables: soda, fries and entrée prices, and registers
- **8 cells, 147 lines, no comments**
- A dataclass, a regular expression, `methodcaller`, `product`, styled tables, a histogram
- Runs top to bottom with no errors

COMPLETE?

Yes. Every step runs and every table prints.

CORRECT?

One thing in it is wrong. The rounds find it.

AFTER THE NOTEBOOK

Complete, and now understood

COMPLETE

- Runs, and prints every table
- 147 lines, none explained
- Standard errors too small in every row

COMPLETE AND UNDERSTOOD

+ Runs, and prints every table
+ Every line explained
+ Standard errors use the N in the table

Both versions run. Only the second is correct, and only the second can be explained.

BEFORE YOU START

Write the rules into your README

YOU ASK

> Add a section to README.md called “Our rules for working with AI” with tonight's rules. Show me before you save.

> Read README.md and follow its rules while we work.

WHY THE README

- The rules live in one place, so you don't retype them
- Start each session by asking the assistant to read it
- Add a rule whenever you learn one
- Read what it wrote before you keep it — the rules are yours

YOUR TURN

Working through Problem Set 1

AS YOU WORK

- Iterate on anything you don't understand
- Don't move on until you can explain it
- What a result means is yours to stand by

WHAT YOU TURN IN

- Simple, straightforward code, annotated line by line

PROMPTS TO START FROM

- “Walk me through this cell. Don't change anything.”
- “Rewrite it the simplest way that gives the same output.”
- “Comment every line for someone who has never coded.”
- “What does this assume about missing values?”
- “How can I check this number is right?”

PROBLEM SET 1

What to check before you trust a number

AGAINST THE DOCUMENTATION

- The problem set says **473** observations; `public.dat` has **410**
- FTE, % full-time and full-meal price are not in the file. The definitions are in the paper
- Every row has its own **N**, and the standard error should use it
- If a number differs from Table 2, find out why before changing code

DEFAULTS TO KNOW ABOUT

- **Missing values:** `.mean()` skips them without saying so
- **Section III:** set a seed. One draw can mislead — in the demo, one draw of classification error came out with the wrong sign

OPEN FLOOR

Your Problem Set 1 questions

BRING

- The cell, and the error or output
- What you expected, and what you got
- We talk about approach, not answers
- Stuck later: `jr120@rice.edu`

BEFORE YOU LEAVE

- Your README has tonight's rules; **`decisionlog.md`** has what you changed
- Problem Set 1 is due **Thu Sep 17, 11:59 PM**, on Canvas

NEXT WEEK · GITHUB

A place online that keeps saved versions of your project folder. Have your GitHub account ready.